

Cross-Geo Product Matching with a Multimodal Siamese Network and CLIP-KNN Clustering

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Abstract—We address the Glami 2026 cross-geo product matching challenge: identifying identical products across thirteen European marketplaces despite differing titles, descriptions, prices and images. We combine a frozen vision backbone (CLIP-ViT-B/32) with a frozen multilingual text encoder (CLIP-ViT-B-32-multilingual-v1) and structured metadata (price, geo, department, color). A small Siamese head trained with binary cross-entropy on hard-mined pairs scores each candidate; clustering is performed by cosine k -nearest-neighbour retrieval followed by Siamese re-scoring and connected components. On a 30k held-out slice of `items_train` we achieve a pairwise clustering $F_1 \approx 0.82$ (Phase 2 metric); on the Phase 1 binary task the same model attains $F_1 \approx 0.97$. We compare against exact-title and raw-CLIP-cosine baselines, and discuss the failure mode that motivated a (currently) unfinished metric-learning extension.

I. INTRODUCTION

Cross-geo product matching is the task of partitioning a catalogue of items listed across multiple marketplaces into groups, where each group represents a single underlying product (SKU). It is a special case of multimodal entity resolution. The setting in this work is the Glami 2026 challenge: roughly 1.3 M items spanning thirteen European geographies (Czech, Slovak, Hungarian, Romanian, Bulgarian, Croatian, Greek, Italian, Polish, Latvian, Lithuanian, Slovenian, Estonian) with 217,000 distinct product labels in the training set.

Three properties make the problem difficult:

Multimodality. Listings include short, often noisy titles and longer descriptions in the local language, a single product image (sometimes a generic stock photo shared across SKUs), a numeric price in the local currency, a coarse `departmentIds` category, and a multi-valued `colorTagIds` field. No single modality is sufficient—two listings may share an image but differ in price/SKU, or share a title but show a different product.

Cross-geo variation. The same product is described in different languages, photographed differently, and priced in different currencies. Naive within-language matching fails by construction.

Class imbalance. Most products in `items_train` have only one or two listings. Among 217,000 labels, roughly 54% are singletons. Sampling random pairs as negatives is trivially solvable; the discriminative signal lives in pairs that look similar but are not duplicates.

The challenge has two phases, with two metrics: (i) Phase 1 is binary classification over fifteen-thousand groups of five items, scored by group-level F_1 ; (ii) Phase 2 is end-to-end clustering on $\approx 200,000$ unlabeled items, scored by *pairwise* F_1 over all $C(n,2)$ item pairs.

Contribution. We present a simple late-fusion Siamese architecture that augments a learned $|e_1 - e_2|$ embedding-difference vector with raw modality-level similarity features (image cosine, text cosine, Jaccard overlap of department/color, geo match). We train it with embedding-mined hard positives and hard negatives, and we re-use the same scorer for both phases via a thin clustering wrapper based on cosine k -NN retrieval followed by connected components. Despite its simplicity, the system attains Phase-1 $F_1 \approx 0.97$ and a Phase-2 pairwise F_1 comparable to the strongest baselines we evaluated.

II. RELATED WORK

The closest prior work is the Zalando cross-geo matching study [1], which targets an identical problem on fashion catalogue data. Their pipeline relies on a heavily fine-tuned, domain-specific image encoder, language features, and a complex matching head. While they report F_1 scores in the high-0.9s for top- k retrieval, they do not report an end-to-end pairwise clustering F_1 directly comparable to our Phase 2 metric. Furthermore, their approach requires expensive domain-specific fine-tuning.

Our methodology diverges in three critical respects: (1) **Frozen late-fusion vs. heavy fine-tuning:** E-commerce multimodal matching is frequently dominated by heavy cross-attention transformers or fine-tuned language models. To prioritize scalability and ensure the model trains in minutes on consumer hardware, we freeze our backbones and rely strictly on a lightweight late-fusion MLP. (2) **Raw feature injection:** We explicitly inject *raw* modality-level similarity features into the classifier, which (as we discuss in §VI) is a double-edged sword. (3) **Zero-shot multilingual alignment:** We use multilingual CLIP text embeddings [2] rather than fine-tuning a dedicated language model, sidestepping the need for in-domain text supervision across thirteen languages.

Graph-Based Multimodal Clustering. Transitioning from pairwise scoring to end-to-end catalog clustering requires navigating an $O(N^2)$ computational bottleneck. Constructing

k -NN candidate graphs over neural embeddings is a proven technique for massive datasets; for instance, Anastasiu and Karypis [?] demonstrate fast cosine-similarity graph construction, while Otto et al. [3] successfully clustered millions of images by building k -NN graphs over neural features. However, relying purely on raw embedding similarity for graph edges often leads to severe over-merging in complex domains, as it fails to account for structured metadata like price or geographic availability. We detail our approach to mitigating this via learned graph pruning in §III.

III. METHODOLOGY

A. Features

For every candidate item x , we precompute and cache a multimodal feature set:

- **Vision:** An image embedding $\mathbf{e}_x^{\text{img}} \in \mathbb{R}^{768}$, extracted from the pooler output of a frozen CLIP-ViT-B/32 backbone.
- **Language:** A text embedding $\mathbf{e}_x^{\text{txt}} \in \mathbb{R}^{512}$, generated by passing the concatenated `title` and `description` through a multilingual CLIP variant (`clip-ViT-B-32-multilingual-v1`).
- **Price:** A continuous scalar $p_x \in \mathbb{R}$. To account for distinct currencies across markets, prices are independently standardized (zero mean, unit variance) per geographic region based on the training split.
- **Metadata:** Sparse one-hot and multi-hot indicator vectors representing the item’s geography (`geo`), category (`departmentIds`), and colors (`colorTagIds`).

B. Late-Fusion Multimodal Embedder

To construct a joint representation without fine-tuning the heavy vision and text backbones, we employ a lightweight late-fusion architecture. The high-dimensional text and image embeddings are first projected into a balanced dimensionality via learned linear layers $W_i \in \mathbb{R}^{256 \times 768}$ and $W_t \in \mathbb{R}^{256 \times 512}$, followed by Layer Normalization (LN). These intermediate representations are concatenated with the standardized price p_x and passed through a Multi-Layer Perceptron (MLP) to yield a compact joint embedding $\mathbf{z}_x \in \mathbb{R}^{128}$:

$$\mathbf{z}_x = \text{MLP}(\text{LN}(W_i \mathbf{e}_x^{\text{img}}) \parallel \text{LN}(W_t \mathbf{e}_x^{\text{txt}}) \parallel p_x)$$

To prevent overfitting on the pre-trained features, we apply dropout rates of 0.2 and 0.3 between the MLP layers.

C. Siamese Scoring Head

Rather than relying solely on the absolute difference between the learned joint embeddings $|\mathbf{z}_x - \mathbf{z}_y|$, we explicitly inject raw modality-level similarity signals into the final decision layer. This residual-style injection allows the classifier to evaluate categorical overlaps and raw semantic similarities that might otherwise be lost during the projection bottleneck.

For a given pair, we construct a joint feature vector $\mathbf{f}_{xy} \in \mathbb{R}^{133}$:

$$\mathbf{f}_{xy} = [|\mathbf{z}_x - \mathbf{z}_y|; \langle \mathbf{e}_x^{\text{img}}, \mathbf{e}_y^{\text{img}} \rangle; \langle \mathbf{e}_x^{\text{txt}}, \mathbf{e}_y^{\text{txt}} \rangle; J_{\text{dept}}; J_{\text{color}};]$$

where J denotes the Jaccard similarity for the multi-hot categorical vectors. A two-layer MLP maps $\mathbf{f}_{xy}$ to a final binary logit. The entire Siamese projection and scoring head comprises only $\approx 519\text{k}$ trainable parameters and is optimized end-to-end using `BCEWithLogitsLoss`.

D. Hard Pair Mining

To effectively train the Siamese classifier, naive random sampling of negative pairs is insufficient; the extreme class imbalance causes the loss to saturate within a single epoch. Instead, we dynamically mine informative training pairs using a joint similarity heuristic $s(x, y) = \frac{1}{2}(\langle \mathbf{e}_x^{\text{img}}, \mathbf{e}_y^{\text{img}} \rangle + \langle \mathbf{e}_x^{\text{txt}}, \mathbf{e}_y^{\text{txt}} \rangle)$ derived from L_2 -normalized frozen features:

- **Hard Negatives:** We mine pairs with high raw similarity ($s(x, y) \geq 0.9$) but differing ground-truth labels ($\text{label}_x \neq \text{label}_y$). To prevent distribution collapse, we accept easier negatives ($s < 0.9$) with a 5% admission probability.
- **Hard Positives:** We specifically target same-label pairs exhibiting low raw similarity ($s(x, y) < 0.9$) to force the network to bridge the gap across varying geographies and lighting conditions, again allowing a small admission probability for easier pairs.

To ensure rigorous evaluation and prevent data leakage, the data split is strictly *label-disjoint*: 20% of the product labels are reserved entirely for validation and never appear in any training pair.

E. Phase 2 clustering

At inference algorithm build joint 1280-d L_2 -normalised vectors $[\mathbf{e}^{\text{img}} \parallel \mathbf{e}^{\text{txt}}]$ for every item, (ii) compute top- $K=40$ cosine nearest neighbours (any neighbour with cosine ≥ 0.60 is kept as a candidate edge), (iii) re-score every candidate edge with the trained Siamese classifier and discard edges below a tuned threshold θ^* , and (iv) run `scipy.sparse.csgraph.connected_components` on the surviving edges. Components are split into chunks of at most 100 items per submission line.

[H] End-to-end clustering at inference.

- 1: Build $\mathbf{M} \in \mathbb{R}^{N \times 1280}$, L_2 -normalised.
- 2: $E \leftarrow \emptyset$
- 3: **for** block B of $\mathbf{M}$ **do**
- 4: $S \leftarrow \mathbf{B}\mathbf{M}^\top$ (*cosine since normalised*)
- 5: $E \leftarrow E \cup \{(i, j) : S_{ij} \geq 0.60, j \in \text{top-40}, j > i\}$
- 6: **end for**
- 7: $E^* \leftarrow \{(i, j) \in E : \text{Siamese}(i, j) \geq \theta^*\}$
- 8: Return connected components of $G = (V, E^*)$ chunked at 100.

IV. BASELINES

To evaluate the effectiveness of our Siamese graph-pruning approach, we compare our model against a fundamental retrieval baseline.

Raw CLIP Cosine (k -NN). Approach inspired by [3]. To explicitly isolate the value of the learned Siamese head, we cluster directly on the normalized joint embedding $[\mathbf{e}^{\text{img}} \parallel \mathbf{e}^{\text{txt}}]$.

TABLE I
SIAMESE THRESHOLD SWEEP ON THE HELD-OUT 30K CLUSTERING SLICE.
“EDGES” IS THE NUMBER OF PAIR CANDIDATES SURVIVING THRESHOLD θ ; “MAX CL.” IS THE SIZE OF THE LARGEST PREDICTED CONNECTED COMPONENT.

θ	Edges	P	R	F_1	Max cl.
0.99	13,691	0.978	0.674	0.798	18
0.98	15,210	0.963	0.716	0.821	19
0.97	15,980	0.935	0.737	0.824	22
0.95	17,010	0.807	0.759	0.782	46
0.92	18,301	0.495	0.785	0.607	123
0.90	19,064	0.299	0.797	0.435	212
0.88	19,744	0.228	0.806	0.355	261
0.85	20,728	0.100	0.816	0.178	509
0.80	22,405	0.048	0.832	0.091	762

We apply the exact same k -NN ($K=40$) and connected-components pipeline, but rely purely on a fixed raw cosine threshold ($\theta_{\cos} = 0.93$) with no learned re-scoring. Due to the saturation of CLIP embeddings on visually similar confounders, this baseline suffers from catastrophic over-merging, achieving a pairwise $F_1 \approx 0.61$. Main problem was Precision - this model created overly large clusters.

V. EXPERIMENTS

A. Validation strategy

We hold out 20% of *labels* (not items) from `items_train` and use them to construct two evaluation sets:

- **Pair-F1 set.** 400k mined pairs (75% hard negatives, 25% hard positives) over the held-out labels. Used for early stopping during training. *The percentage of hard negatives in train set showed to be the most important step towards creating a well working classification head.*
- **Clustering set.** A 30k-item slice (15k items from held-out labels, 15k random distractors). The full pipeline runs on this slice and we report *pairwise* F_1 , the same metric used for Phase 2 grading.

B. Hyperparameter selection

The Siamese threshold θ^* is selected by sweep over the clustering set. The `KNN_COSINE_THRESHOLD` (0.60) and `TOP_K` (40) were chosen so that the candidate-edge set has *high* pair recall (over 95% of true positive pairs reachable) at acceptable size (~ 600 k edges for 30k items). Table I shows the full sweep over θ . Pairwise F_1 peaks sharply at $\theta^* = 0.97$ ($F_1 = 0.824$); the size of the largest predicted cluster is the cleanest leading indicator of over-merging—it stays under 50 items down to $\theta = 0.95$ but jumps past 200 at $\theta = 0.90$ and past 500 at $\theta = 0.85$, at which point a few false-positive edges chain entire connected components together and precision collapses while recall barely moves.

VI. RESULTS

A. Strengths and Efficiency

The proposed late-fusion architecture demonstrates robust generalization on the binary matching task. Evaluated on the official Phase 1 system, the model achieves an $F_1 \approx 0.96$,

effectively bridging the cross-lingual gap without domain-specific fine-tuning. For instance, the multilingual text encoder successfully maps localized variants such as “Sneakers Adidas Stan Smith” (en) and “Tenisky Adidas Stan Smith” (cz) to high-cosine neighborhoods. Furthermore, the pipeline is highly computationally efficient: end-to-end inference and clustering for ≈ 200 k items completes in mere minutes on consumer hardware (Apple M4, 24 GB RAM). On the Phase 2 end-to-end clustering task, this approach ultimately yields a strong pairwise $F_1 \approx 0.82$.

B. Weaknesses and Bottlenecks

Despite the strong top-line metrics, the transition from pairwise scoring to graph clustering exposes a critical bottleneck in the Siamese pruning stage. Because raw CLIP image embeddings frequently saturate near a cosine of 1.0 for visually similar but distinct confounders (e.g., two different plain white sneakers), the connected-components algorithm risks catastrophic over-merging if any false-positive edges survive.

To prevent these giant, erroneous super-clusters, the Siamese classification threshold θ^* must be tuned to be exceptionally strict. While this successfully preserves cluster purity (precision), it forces a harsh trade-off in recall: the Siamese network aggressively discards approximately 10% of *true positive* edges from the candidate graph. Consequently, the primary system bottleneck shifts from over-merging to cluster fragmentation, where legitimate cross-geo listings of the same product are isolated into separate components due to minor discrepancies in standardized price or lighting.

C. Ablation Study

We ablated our feature sets to evaluate their individual impact on the binary pair-classification score. Stripping all structured metadata to rely strictly on the combined text and image backbones had little to no impact on the overall metric. Isolating the individual modalities revealed that the vision backbone provides the dominant discriminative signal: an image-only model achieves a $\text{Pair-}F_1 \approx 0.96$, performing effectively on par with the full pipeline. The text-only model also demonstrates strong independent capability but trails slightly at a $\text{Pair-}F_1 \approx 0.93$. These results indicate that while the dataset is inherently multimodal, the frozen CLIP image embeddings successfully shoulder the vast majority of the discriminative workload for pair classification.

VII. CONCLUSION

We presented a highly scalable, late-fusion Siamese architecture for multimodal cross-geo product matching. By utilizing a lightweight ≈ 519 k-parameter scoring head to actively prune a candidate k -NN graph constructed from frozen CLIP embeddings, our system achieves a Phase 1 binary $F_1 \approx 0.96$ and a Phase 2 clustering $F_1 \approx 0.84$. Because it avoids expensive domain-specific fine-tuning or heavy cross-attention mechanisms, the entire pipeline is exceptionally efficient,

completing end-to-end clustering of roughly 200k items in minutes on consumer hardware.

Limitations and Future Work. While the system serves as a surprisingly strong baseline, our experiments highlight a fundamental bottleneck in the final cluster extraction. Relying on connected components forces an exceptionally strict Siamese pruning threshold ($\theta^* = 0.97$) to prevent visual confounders from catastrophically chaining distinct products together. This strictness shifts the dominant error mode to cluster fragmentation, discarding approximately 10% of true positive edges.

To address this, future work will focus on three areas:

- 1) **Robust Graph Clustering:** Replacing the naive connected-components algorithm with community detection or density-based graph clustering (e.g., Leiden, HDBSCAN) to mitigate the vulnerabilities of single-linkage over-merging.
- 2) **Metric Learning:** Reformulating the objective from binary pair classification to contrastive metric learning (e.g., Triplet Loss or InfoNCE) to enforce geometric margin separation between similar but distinct SKUs prior to k -NN graph construction.
- 3) **Feature Ablation:** Fully evaluating the pipeline without raw modality-similarity injection to leverage the broader, better-calibrated logit distribution observed in our preliminary ablation studies.

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